SMART EXPENSE TRACKER ALGORITHM

Step 1: Start

- Begin the program.

Step 2: Login / Create Password

- Prompt the user to enter a password.
- If the user is new, allow them to create and save a password.

Step 3: Validate Password

- Check whether the entered password is correct.
- If correct → proceed to menu.
- If incorrect → display error message.
- Allow only 3 attempts.
- If attempts exceed limit → terminate the program.

Step 4: Display Menu

- Show the following options:
    - Add Expense
    - View Expenses
    - Search by Date
    - Monthly Report
    - Total Expense
    - Edit Expense

- Delete Expense
- Category-wise Total
- Change Password
- Exit

Step 5: Perform Selected Operation

Add Expense

- Input date, category, and amount.
- Save the data in expenses.csv.

View Expenses

- Read all records from expenses.csv.
- Display all stored expenses.

Search by Date

- Input a specific date.
- Find and display matching records.

Monthly Report

- Input month (MM).
- Calculate total expenses for that month.
- Display the result.

Total Expense

- Read all expense amounts.
- Calculate total sum.
- Display total expense.

Edit Expense

- Input date or record identifier.
- Modify required details.
- Update the record in the file.

Delete Expense

- Input date or record identifier.
- Remove the selected record.
- Save updated file.

Category-wise Total

- Calculate total expenses for each category.
- Display category-wise totals.

Change Password

- Verify old password.
- Enter and save new password.

Exit

- Terminate the program.

Step 6: Return to Menu

- Ask the user whether to return to the menu:
    - If **Yes** → go back to menu
    - If **No** → proceed to end

Step 7: End

- Stop the program execution.